AIN SHAMS UNIVERSITY-

FACULTY OF ENGINEERING

[MCT312s]

Industrial Automation

Team 1

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Responsibility sheet and task allocation for each team member

NO	Name	Task	Percentage %
1	Abdelkarim Wael Abdelkarim	-Circuit Design 2D -Purchasing the components -Circuit Connection	19.5
2	Mahmoud Salama Mohamed Abdo	-Circuit 3D Modelling -Report -Connection -Labelling	16
3	Musa Gamal El-Shenawy	-Report -Purchasing the components -Circuit Connection	17
4	Abdullah Mahmoud Abdelkarim	-Circuit 3D Modelling -Purchasing the components -Circuit Connection	19.5
5	Youssef Ayman	-Circuit 3D Modelling -Labelling	14
6	Sam Isaac	-Circuit 3D Modelling -Labelling	14

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acronym list

Symbol	Description
NO	Normally Open
NC	Normally Closed
P1(LS1)	Limit switch 1 For Fully Retracted
P2(LS2)	Limit switch 2 For Fully extended

Introduction

Industrial automation is the application of control systems, like computers and robots, to handle processes and machinery in industrial settings. This technological shift significantly enhances efficiency, accuracy, and productivity in various sectors.

Key Benefits of Industrial Automation:

- **Increased Efficiency:** Automation reduces labour costs and increases production speed.
- **Improved Quality:** Consistent product quality and reduced defects.
- **Enhanced Safety:** Minimizes human error and reduces risks in hazardous environments.
- **Optimized Resource Utilization:** Maximizes the use of resources like energy and materials.

Industrial automation can be categorized into three primary types:

Programmable Automation: Highly flexible, uses PLCs to control processes.

Flexible Automation: Balances flexibility and efficiency, employs robots and AGVs.

Fixed Automation: Designed for high-volume, repetitive tasks, uses hardwired control systems.

In this project we will focus on hardwired automation:

Hardwiring refers to the physical connection of electronic components using fixed circuits. In the context of automation systems, hardwiring involves using cables and connectors to establish a permanent connection between sensors, actuators, and control units. While hardwiring offers reliability and deterministic performance, it can limit flexibility and scalability. Understanding the intricacies of hardwiring is crucial for designing, implementing, and maintaining automation systems.

Project Summary

- This project focuses on simulating and operating a control panel for a hydraulic cylinder system.
- The controlling panel manages the extending and retracting a hydraulic piston cylinder with 2 limit switches.
- The primary objective is to understand of the panel's components, their interactions, and the role of hardwiring in its operation.

Specific Objectives:

1. **Component Identification:** identify and understand the function of each component, including the hydraulic cylinder, valves, pump, sensors, and control logic.

2. **Hardwiring Analysis:** Analyse the hardwiring scheme, focusing on the connections between components, signal flow, and the role of specific wires.

3. **Control Logic Simulation:** Simulate the panel's control logic and analyse its behaviour.

4. **Real-World Panel Construction:** Build a physical control panel that can be integrated it with the hydraulic system.

Within the project we will focus on the components and the automation circuit.

system scenario layout:

A double-acting cylinder is used to seal jar caps. Once the cylinder completes its sealing cycle, an electrical relay is activated to start the conveyor motor, moving the next jar into position

Layout:

- Conveyor System: Transports jars to the sealing station.
- Double-acting cylinder: Seals the jar caps.
- Limit Switch: Detects the completion of the sealing cycle.
- Electrical Relay: Connects the cylinder's retraction circuit to the conveyor's start circuit.

Description:

1. Jar Arrival: A jar is positioned under the sealing head.
2. Cylinder Extension and Sealing: The cylinder extends and seals the jar cap.
3. Cylinder Retraction and Relay Activation: Once the sealing is complete, the cylinder retracts, triggering the limit switch.
4. Conveyor Movement: The activated relay energizes the conveyor motor, starting the conveyor and moving the next jar into position.
5. Cycle Repetition: The process repeats for the next jar.

Steps For Electrical Control Panel Manufacturing:-

1) **Hardwired Circuit Design and Simulation:** -
   Design and simulate the electrical circuit to ensure optimal performance and safety.
2) **3D Modelling and Wiring Diagram:-**
   Develop the 3D model and wiring diagram of the control panel.
3) **Component Procurement:** -
   Determine and purchase the needed components for the control panel.
4) **Component and Wire Coding:** -
   Code the parts and wires for easy assembly and maintenance.
5) **Panel Fabrication:** -
   Manufacture the panel box according to the 3D model to fix the components in it.
6) **Component Mounting and Wiring:** -
   Mount the components and connect them according to the wiring diagram.
7) **Labelling:** -
   Label all components and wires clearly for easy identification and future maintenance.

Automation circuit :

We used **CADE_SIMU** To Design and Simulate the Automation Circuit.

1st industrial circuit

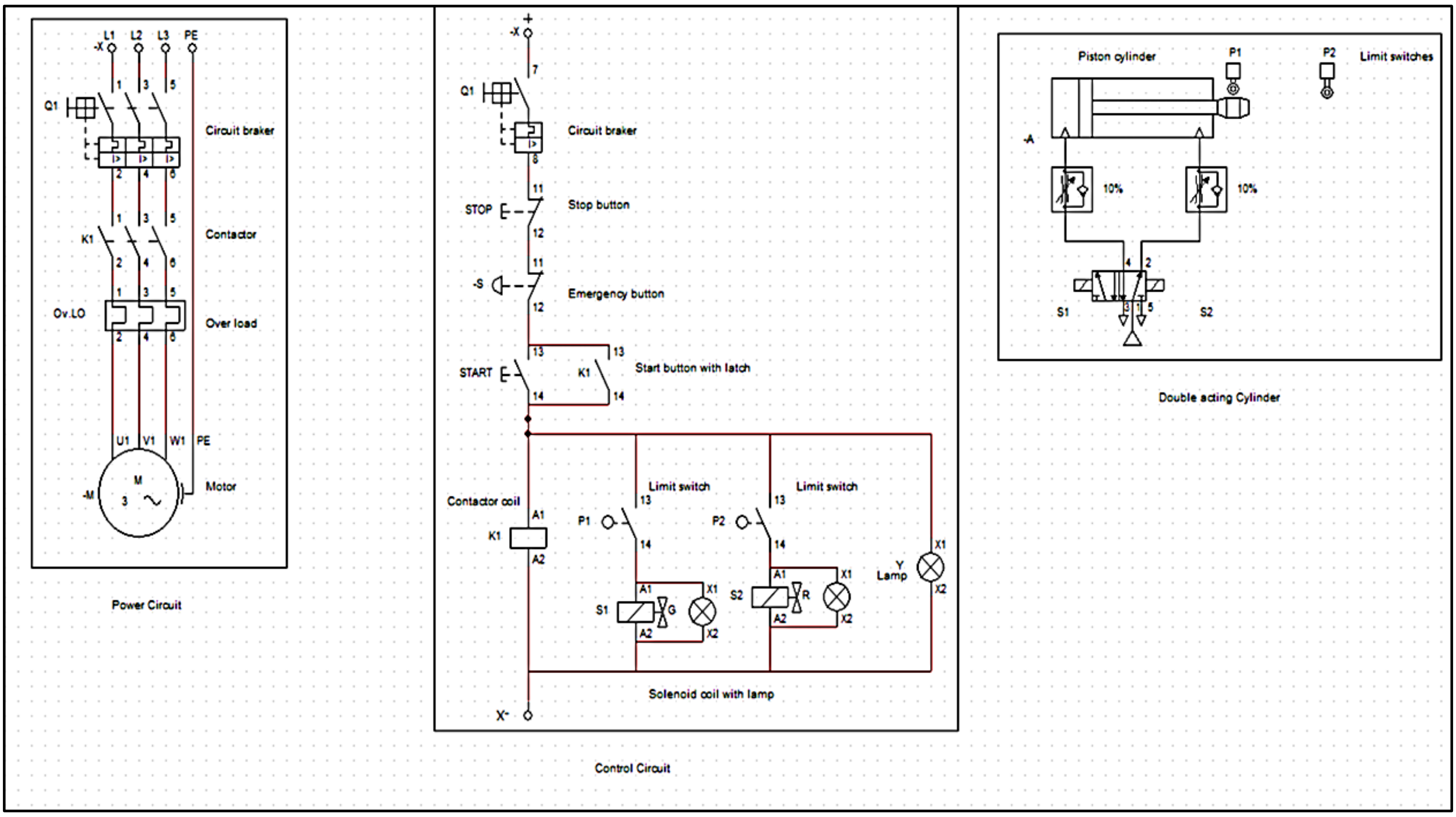


Figure 1: Industrial Circuit

It's the full circuit which includes the control panel and the actuators and the hydraulic system.

When the system starts, the motor-driven pump supplies hydraulic fluid to the cylinder, causing it to extend. The extension continues until the cylinder reaches limit switch P2. Upon activation of P2, the 4/2 solenoid valve switches its orientation, reversing the fluid flow. The cylinder then retracts until it reaches limit switch P1 and This cycle repeats continuously.

Drive Link Simulation of the industrial Circuit:-

https://drive.google.com/file/d/1AXfAVwLhxbN9mvqJFKnlYuV-bltQRuqS/view?usp=sharing

The industrial circuit is divided into 3 circuits:

1- **Power Circuit**
2- **Control Circuit**
3- **Hydraulic Circuit**

Industrial Power circuit.

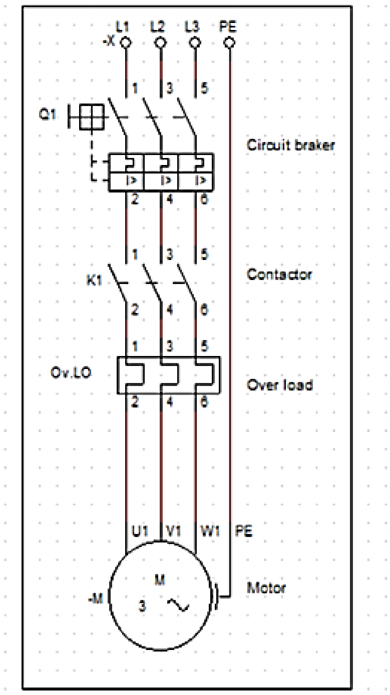


Figure 2 : Industrial Power Circuit

The A power circuit is essential for supplying electrical energy to the control panel and ensures the safety of the system, it may use 3 phases or single-phase power depending on the power regulations of the components.

It consists of:

- **Circuit Breaker:** Protects the circuit from overcurrent and short circuits.
- **Contactor:** A type of relay that switches large electrical loads, such as the motor driving the hydraulic pump.
- **Thermal overload relay** :serves as a crucial protective device in electrical systems, primarily used to safeguard motors from overheating and potential damage due to excessive current. It operates based on the principle of thermal sensing, monitoring the heat generated by the motor during operation.
- **Hydraulic Pump Motor:** Converts electrical energy into mechanical energy to drive the hydraulic pump.

Industrial Control circuit:

A control circuit is the brains behind a hydraulic cylinder system, responsible for interpreting input signals and generating output signals to control the cylinder's movement. usually, it works on a single-phase power supply.

It consists of:

- **Circuit Breaker:** Protects the circuit from overcurrent and short circuits.
- **Stop button:** to stop the system.
- **Emergency button:** to manually stop the system in case of errors or emergency.
- **Start with latch:** button to start the system operation and it latches using the conductor normally opened switch.
- **Limit switches:** Limit switches are electromechanical devices that detect the position of a moving object.

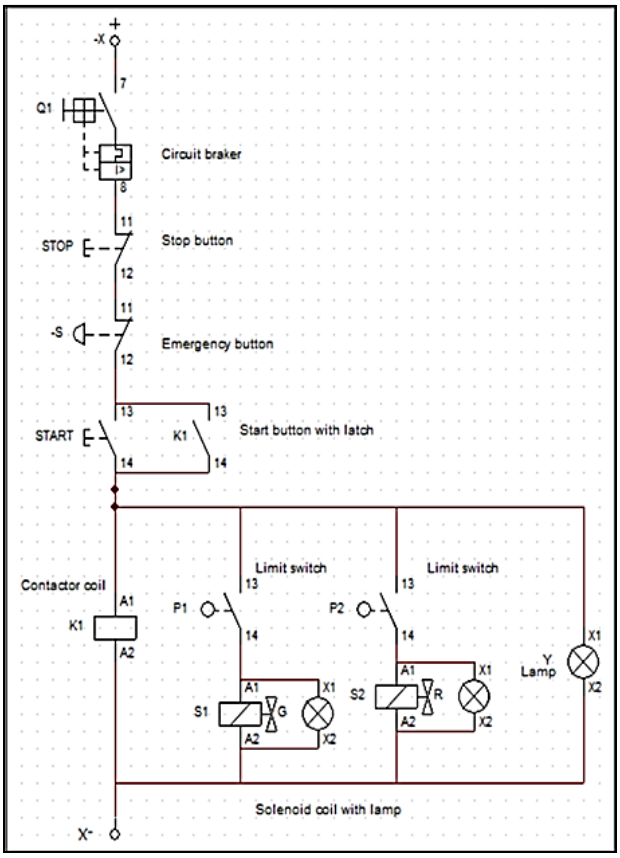


Figure 3: Industrial Control Circuit

hydraulic circuit:

A double-acting cylinder circuit is a common hydraulic system that utilizes a hydraulic cylinder capable of extending and retracting under fluid pressure. A 4/2 directional control valve is typically used to control the fluid flow to the cylinder's ports, enabling bidirectional movement.

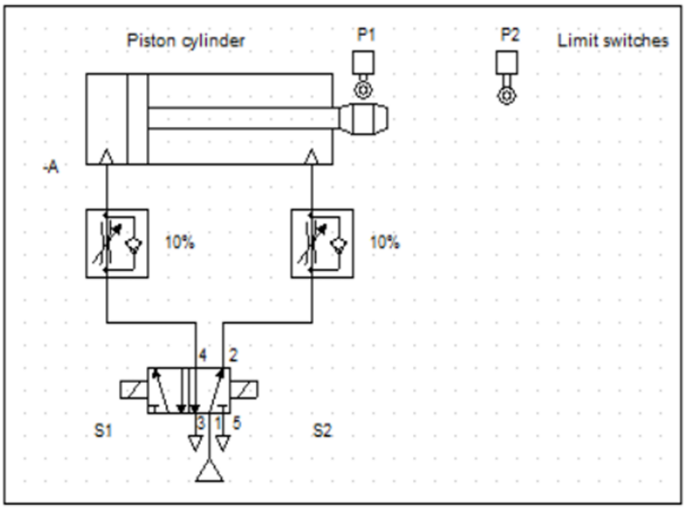


Figure 4:l Hydraulic Circuit

It consists of:

- **4/2cDirectional Control Valve**: Controls the flow of hydraulic fluid to the cylinder's ports, determining its direction of movement. (controlled by solenoid s1)
- **Double-Acting Cylinder**: Converts hydraulic energy into mechanical work.
- **Flow Control Valve:** Regulates the flow rate of hydraulic fluid to the cylinder.

- **Limit switches:** Limit switches are electromechanical devices that detect the position of a moving object.

2nd Control Panel Model Circuit

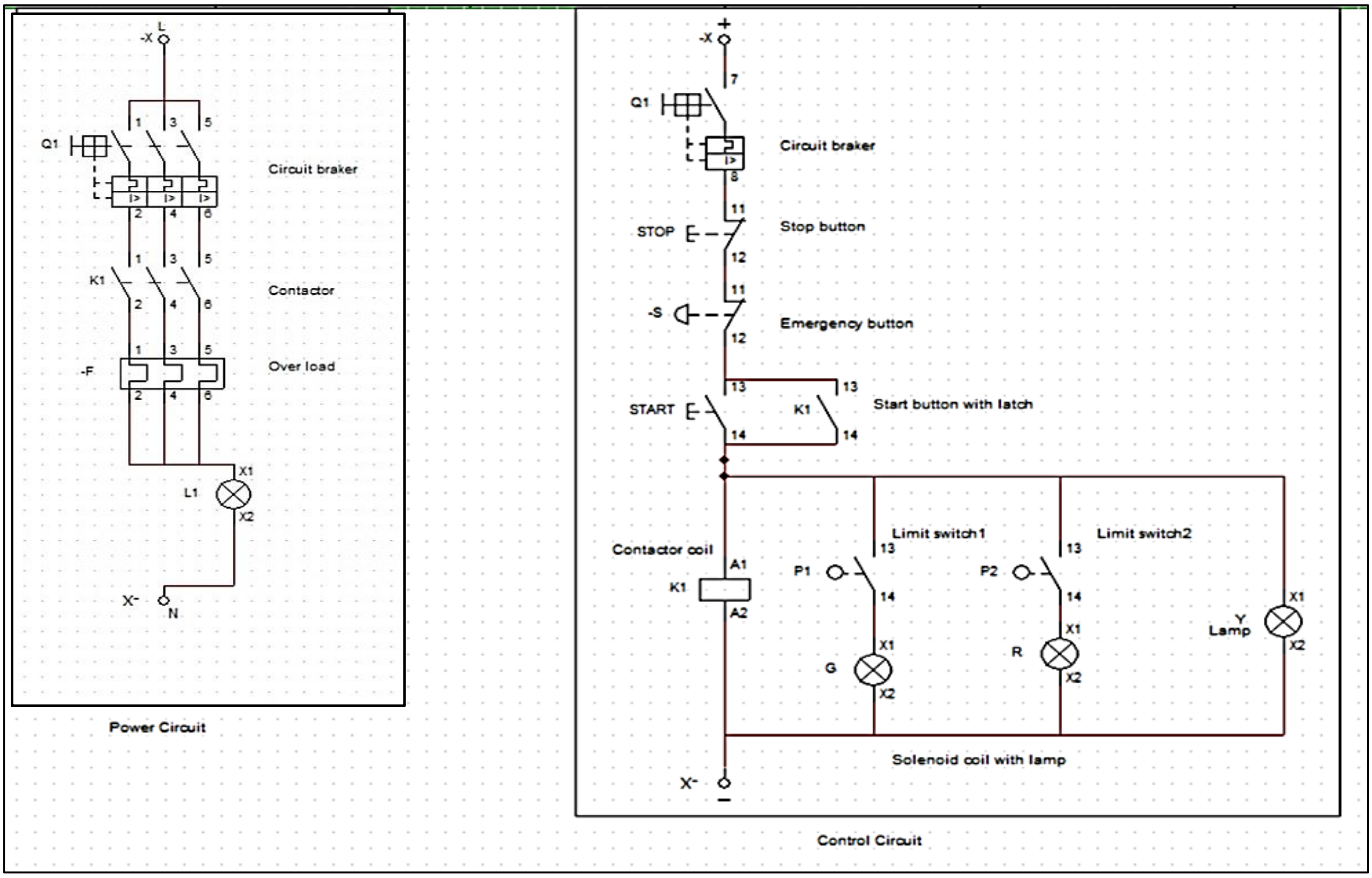


Figure 5: Control Panel Circuit

It’s the simulated model of the industrial control panel which uses lamps as indicators instead of using actuators.

circuit description

when start button pressed it operates contactor relay k1 which latches the start button and connect electricity to the system , when limit switch p1 is pressed it indicates the cylinder is fully retracted and operates green lamp , when limit switch p2 is pressed it indicates the cylinder is fully extended operates red lamp , the yellow lamp indicates that the system is operating

The control panel circuit is divided into 2 circuits:

1-Power Circuit

2-Control Circuit

Power circuit:

It's the same circuit as the industrial circuit but instead of the hydraulic pump motor it uses a lamp as indicator of power.

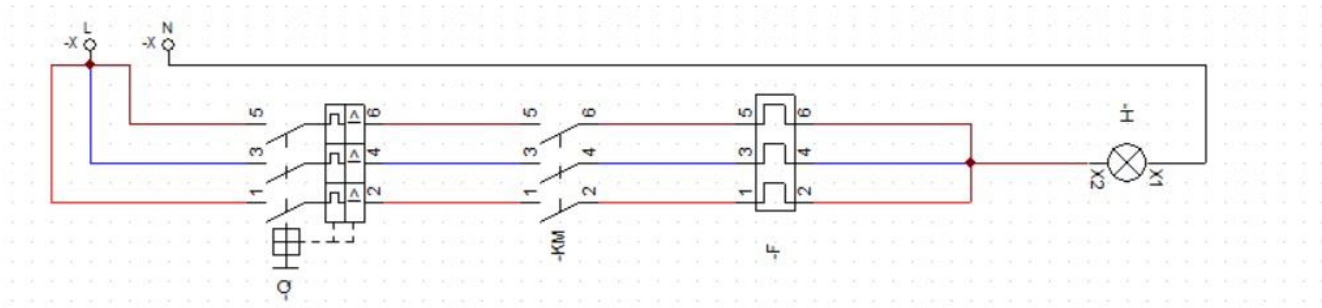


Figure 6: Power Circuit

Control circuit: **(Hardwired Ladder Diagram)**

It's the same control circuit of the industrial circuit but it uses lamps instead of the solenoid valve coils.

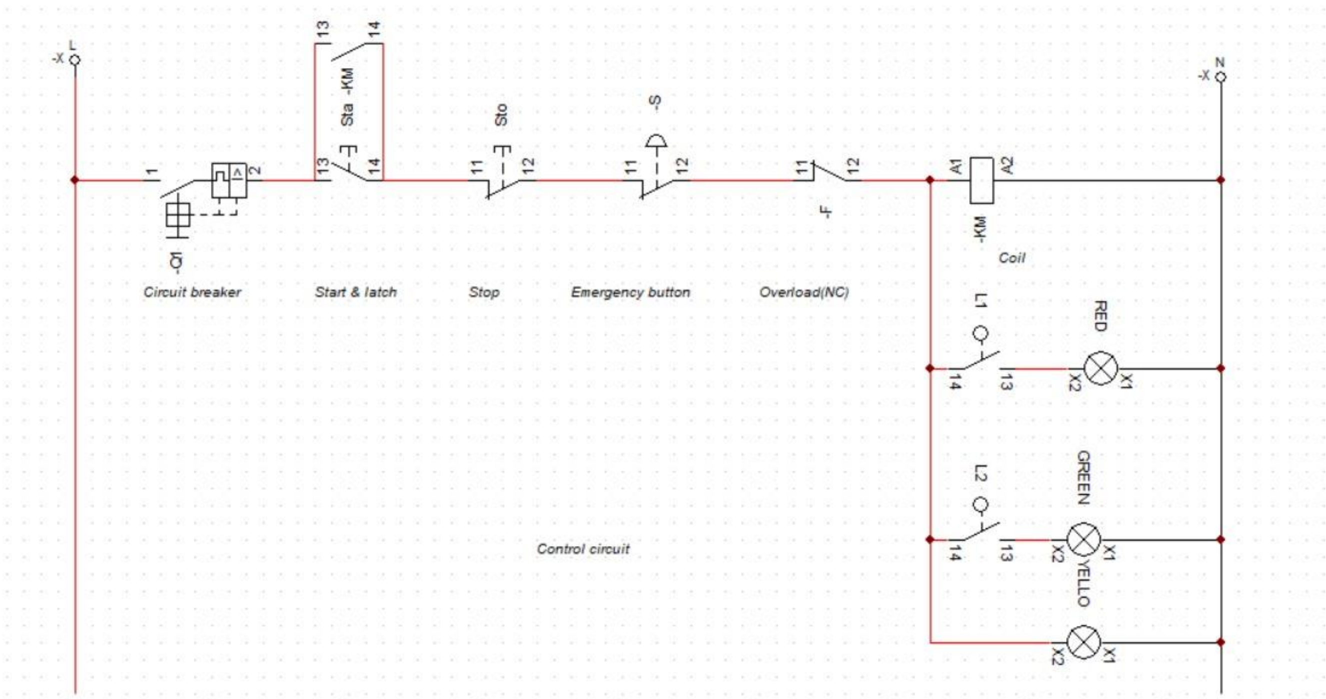


Figure 7: Control Hardwired Ladder Diagram

Drive link Simulation Of the control panel Circuit:

https://drive.google.com/file/d/1VLs7-t8xAOw7XpUtFUocQU1oPCuD5V3J/view?usp=sharing

3D Model and Wiring Diagram:-

We used **SolidWorks_Electrical** To Design and Simulate the 2D and 3D Model and Wiring Diagram

2D Model:-

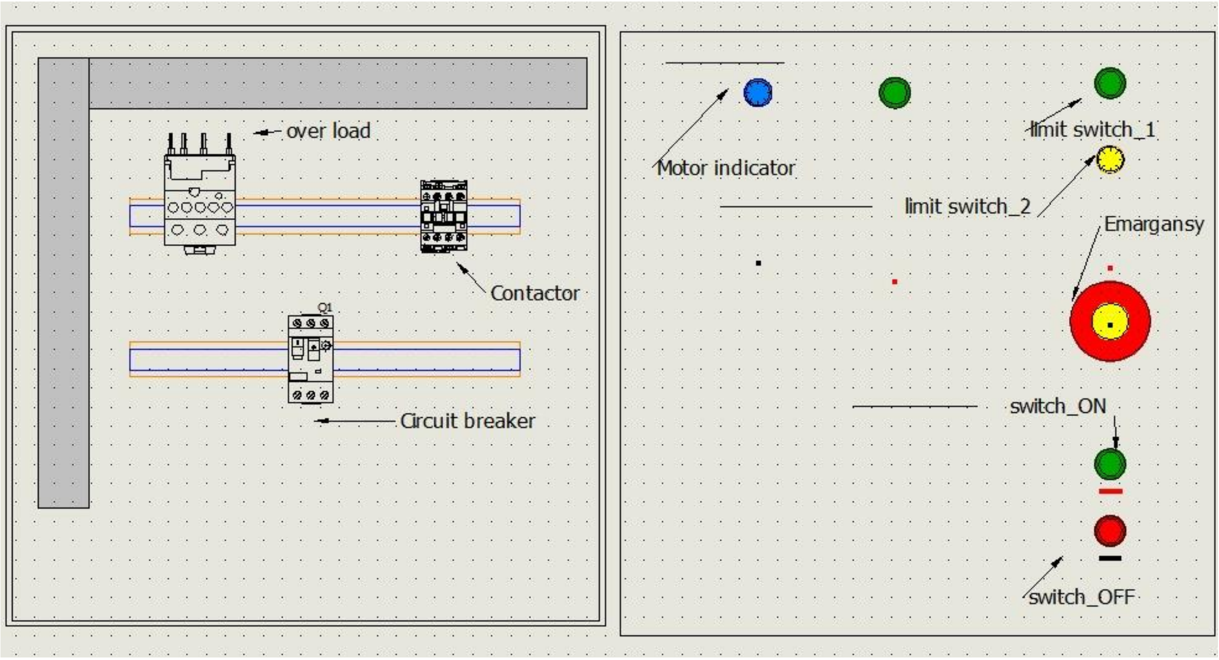


Figure 8: 2D Model

3D Model :-

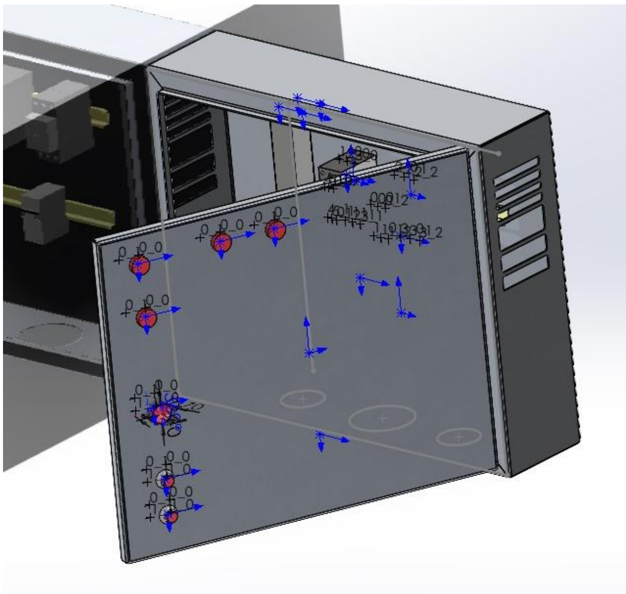


Figure 9: 3D Model Panel Cover

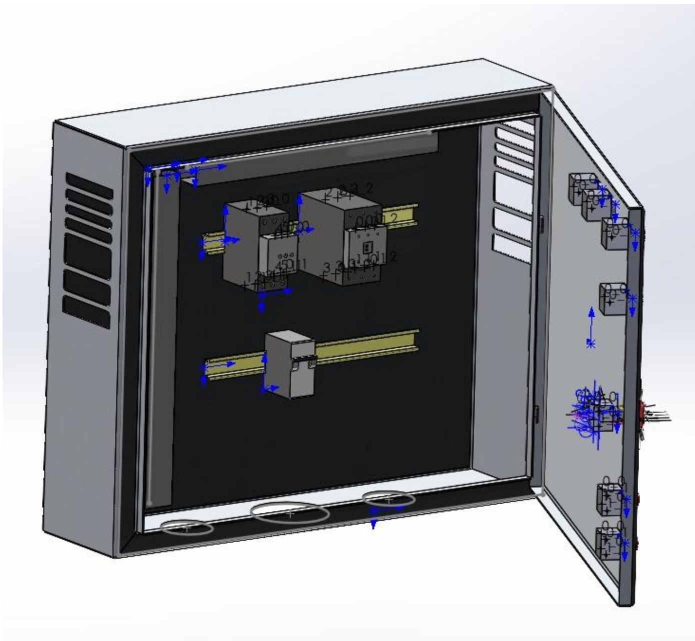
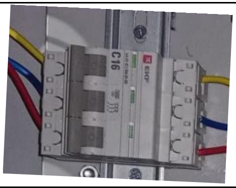
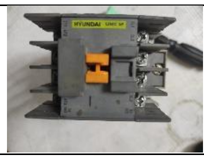
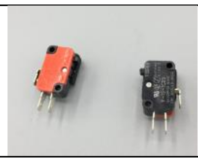


Figure 10: 3D Model inside

Solidworks File:-

https://drive.google.com/file/d/1mkjdzMZ0FjW3Z9sN1BIJG0IgY8ntQR-L/view?usp=sharing

List of Component

No	Name	Role	characteristics	Photo
1	Circuit Breaker	for power circuit to protect all circuit components from over current	3phase 340 v 16 A	
2	Circuit Breaker	for control circuit to protect circuit components from over current	1 phase 220-240 v 10 A	
3	Contactor	it used to control motor on/off by using coil and NO contact points	3phase 200-460 v 16 A	
4	Overload Relay M32 https://kentstore.com/content/pdf/MT-32_specs.pdf	to protect motor from over current by heat generated and bimetallic switch	3phase 340 v 16 A	
5	ON,OFF Push buttons	Two switch on and off turn on system or turn off	1 phase 220-240 v 3A	
6	Emergency Stop Button	The function of an emergency stop button is to act as a safety mechanism it stops all functions immediately when it cannot be shut down manually by a user.	220-240 v 3A	
7	Red LED	an indication that piston is fully retracted	220-240 v	
8	Green LED	an indication that piston is fully extended	220-240 v	
9	Yellow LED	an indication that control circuit is working	220-240 v	
10	Blue LED	an indication that the motor is work	220-240 v	
11	Limit Switch LS1, LS2	to control the limits of Expanding and Retracting of the Cylinder	220-240 v 5A	
12	Electrical Panel box , Duct and Wires			

Wiring Selection:-

Wire Diameter:

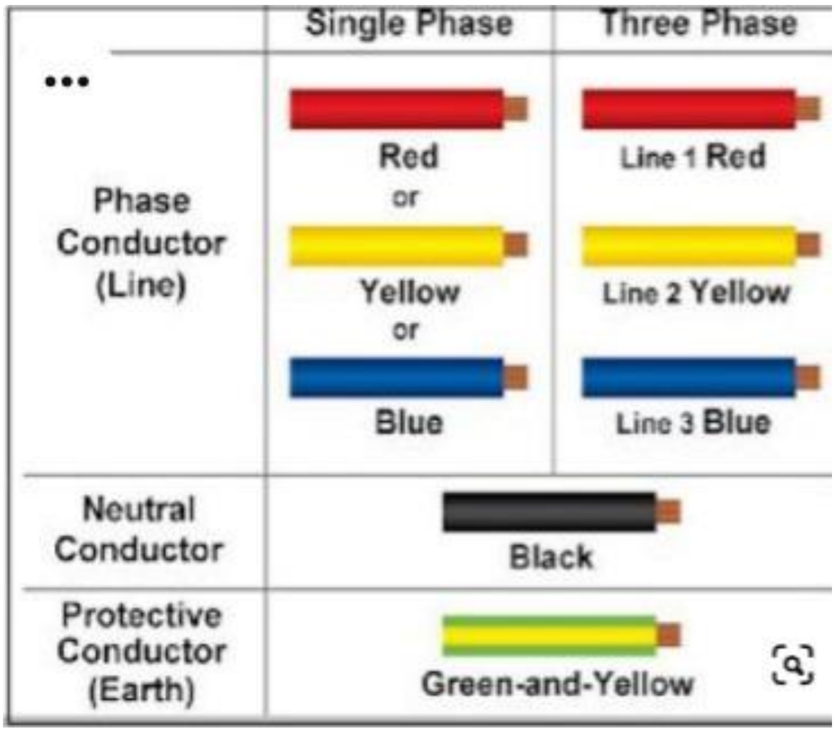
We Choose a motor with Rated current 10.3 A

So we select the wire diameter is 2.3 mm according to American Wire Gauge (AWG) standard.

AWG	Diameter [inches]	Diameter [mm]	Area [mm^2]	Resistance [Ohms / 1000 ft]	Resistance [Ohms / km]	Max Current [Amperes]	Max Frequency for 100% skin depth
0000 (4/0)	0.46	11.684	107	0.049	0.16072	302	125 Hz
000 (3/0)	0.4096	10.40384	85	0.0618	0.202704	239	160 Hz
00 (2/0)	0.3648	9.26592	67.4	0.0779	0.255512	190	200 Hz
0 (1/0)	0.3249	8.25246	53.5	0.0983	0.322424	150	250 Hz
1	0.2893	7.34822	42.4	0.1239	0.406392	119	325 Hz
2	0.2576	6.54304	33.6	0.1563	0.512664	94	410 Hz
3	0.2294	5.82676	26.7	0.197	0.64616	75	500 Hz
4	0.2043	5.18922	21.2	0.2485	0.81508	60	650 Hz
5	0.1819	4.62026	16.8	0.3133	1.027624	47	810 Hz
6	0.162	4.1148	13.3	0.3951	1.295928	37	1100 Hz
7	0.1443	3.66522	10.5	0.4982	1.634096	30	1300 Hz
8	0.1285	3.2639	8.37	0.6282	2.060496	24	1650 Hz
9	0.1144	2.90576	6.63	0.7921	2.598088	19	2050 Hz
10	0.1019	2.58826	5.26	0.9989	3.276392	15	2600 Hz
11	0.0907	2.30378	4.17	1.26	4.1328	12	3200 Hz
12	0.0808	2.05232	3.31	1.588	5.20864	9.3	4150 Hz
13	0.072	1.8288	2.62	2.003	6.56984	7.4	5300 Hz
14	0.0641	1.62814	2.08	2.525	8.282	5.9	6700 Hz
15	0.0571	1.45034	1.65	3.184	10.44352	4.7	8250 Hz

Wire Coding:-

We choose the wire colour according to Standards



Implementation Control Panel:-

Figure 12: Control Circuit

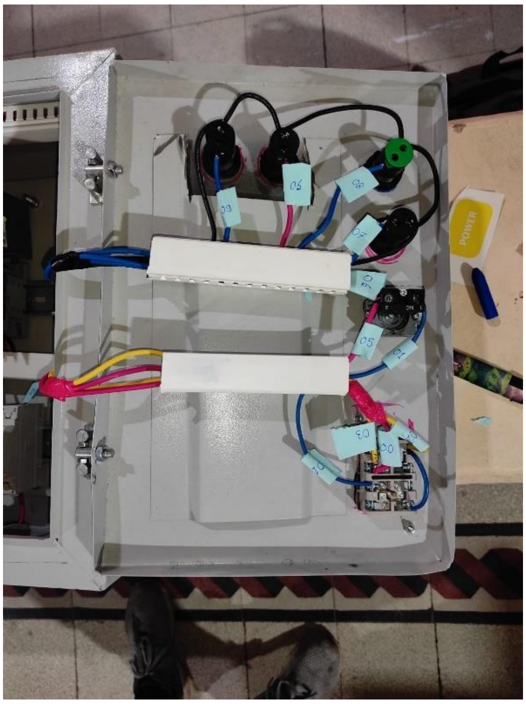


Figure 11: Labelled Control circuit

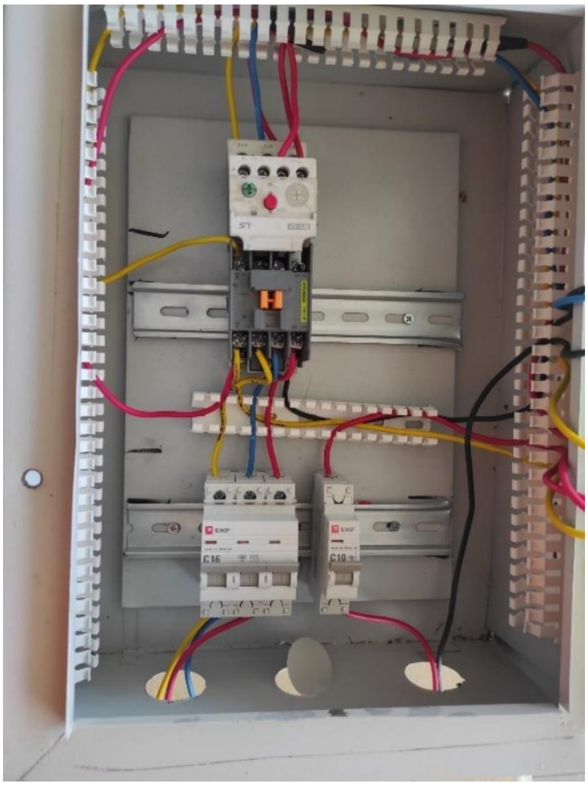


Figure 14: Power Circuit

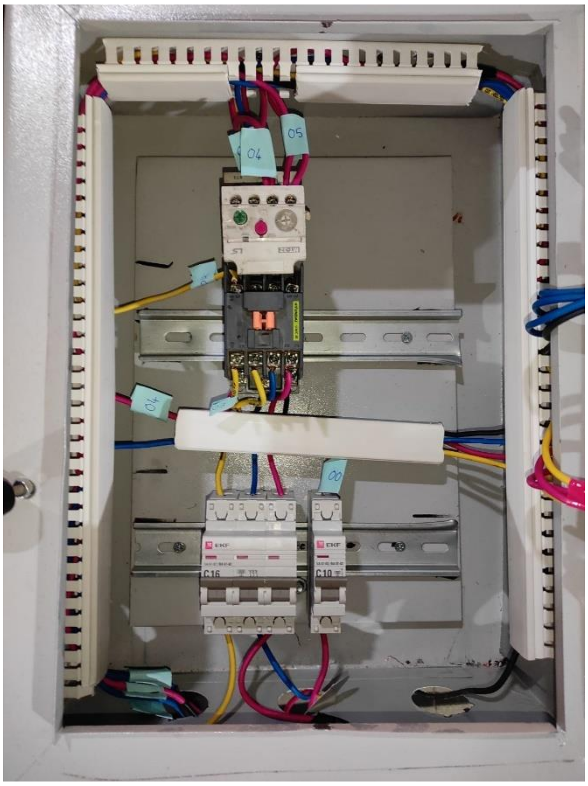


Figure 13: Labelled Power Circuit

Labelling:

label	Description
00	ON-Circuit Breaker
01	Off-Emergency
02	ON-NO contacter
03	ON- NO contactor
04	Limit switch- NC overload relay
05	Emergency- NC overload relay
06	Emergency - Yellow Led
07	Limit switch - Red Led
08	Limit switch - Green Led
09	Overload relay- Blue led

Testing :-

https://drive.google.com/file/d/10U3l7anMuX0P6ioPTWcRQP9o1-E0bxOd/view?usp=sharing

Reference:-

- American Wire Gauge (AWG) standard.
- The NEC wire colour coding standards.